

STAT5020 — Assignment 2 (Part 2: Inference)

Students' Performance Again

Overview

You are provided with a data set containing comprehensive information on **2,392 high school students**, including demographics, study habits, parental involvement, extracurricular activities, and academic performance. The dataset was shared by **Rabie El Kharoua** (2024). “Students Performance Dataset”. A full data description is provided in the Appendix.

Use **R** and **Quarto** to load and analyse the data and answer the questions below.

Part 0 — Understanding the Data (0 marks)

Each data analysis and each report should start with data introduction. To some extent you already completed this task in Assignment 2 Part 1: Data. So, we skip this step for Part 2 of your assignment. Just remember, data introduction should be here for any real report.

This only new thing you must bring for this assessment is a confidence level/significance level. Your data set has a relatively large sample size, so we might expect high confidence level. However, the data are from social science, it is about human behaviour. This type of data tends to have higher variability, that pushes confidence level down. As a result:

For all statistical tests you should use **90% Confidence level**, or respectively **10% Significance Level**.

Part 1. Effect of Parental Support on students' performance measured by GPA (25 marks)

Use the available data to study if there is any effect on students performance by different levels of parental support.

Your answer should include:

- Null and alternative hypotheses
 - Appropriate graphical and numerical summaries with related discussion
 - Correct statistical test with check for its assumptions
 - Results interpretation and conclusion for the research question
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Part 2. Your own research questions (75 marks)

Propose interesting research questions that can be investigated using this data set. For example, as in Part 1 above. And then, you must answer your own research question by using appropriate statistical test.

You must create three different research questions that should be investigated by three different statistical tests. One question for each test.

- t-test (25 marks)
- One-way ANOVA (25 marks)
- Chi-squared test (25 marks)

Your answers for each dot-point above should include:

- Research question formulation
 - Null and alternative hypotheses
 - Appropriate graphical and numerical summaries with related discussion
 - Correct statistical test with check for its assumptions
 - Results interpretation and conclusion for the research question
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It is not expected that you do a lot of writing. In general, there should be:

- One paragraph to introduce the research question and set null and alternative hypotheses.

- One paragraph to discuss presented summaries and report/justify data modifications (if any).
- One paragraph to introduce the test and its assumptions. If some assumptions are violated and it is not really appropriate to use the test for the data, you acknowledge and report them but still proceed with the test.
- One paragraph to interpret statistical test results and make a conclusion with respect to the original research question and overall data context.

Some tests might require a bit more of writing/explanation. It is OK to write more if you need. However, try to be brief as everyone hates reading long reports.

What to submit

Submit **two files** via **myLearning**:

1. A well-presented written report in **HTML or PDF** (rendered from Quarto). Take care that all graphs and tables have meaningful titles, labels, etc.
2. The **Quarto source file** (.qmd) you used to load the data, run the analysis, and produce tables/figures.

Do not submit archived files (zip/rar/7z, etc.).

You **do not** need to submit the data file.

Academic integrity

This is an **individual** assessment. You may discuss general questions and R coding on the forum and in class, but **working in groups is prohibited**.

- You may share ideas, but **not solutions** (neither report text nor R code).
- The use of **generative AI tools (e.g., ChatGPT, Gemini or similar systems)** is strictly prohibited for this assessment.
- By submitting your work, you confirm that it is entirely your own and that you can explain all parts of your submission.
- Any suspected violations will be reported to the Academic Integrity Office for investigation.

If you have questions, ask on the forum or email the teaching team.

Appendix — Data description

Student ID

- **StudentID:** Unique identifier (1001 to 3392).
Note: this is an identifier, not a meaningful variable for analysis.

Demographic details

- **Age:** 15 to 18
- **Gender:** 0 = Male, 1 = Female
- **Ethnicity:**
 - 0: Caucasian
 - 1: African American
 - 2: Asian
 - 3: Other
- **ParentalEducation:**
 - 0: None
 - 1: High School
 - 2: Some College
 - 3: Bachelor's
 - 4: Higher

Study habits

- **StudyTimeWeekly:** 0 to 20 hours per week
- **Absences:** 0 to 30
- **Tutoring:** 0 = No, 1 = Yes

Parental involvement

- **ParentalSupport:**
 - 0: None
 - 1: Low

- 2: Moderate
- 3: High
- 4: Very High

Extracurricular activities

- **Extracurricular:** 0 = No, 1 = Yes
- **Sports:** 0 = No, 1 = Yes
- **Music:** 0 = No, 1 = Yes
- **Volunteering:** 0 = No, 1 = Yes

Academic performance

- **GPA:** 0 to 4
- **GradeClass:**
 - 0: A (GPA ≥ 3.5)
 - 1: B ($3.0 \leq \text{GPA} < 3.5$)
 - 2: C ($2.5 \leq \text{GPA} < 3.0$)
 - 3: D ($2.0 \leq \text{GPA} < 2.5$)
 - 4: F ($\text{GPA} < 2.0$)

Hint: The data is already “encoded”, that is, all categorical variables converted to numbers. This is very common in the industry, and it is OK (as far as we are provided with the full description). However, for some analysis and visualisations it is better to have categorical data as factors. It is very easy to convert numeric data back to categorical. For example, for GradeClass (assuming that the data are stored in dataframe “df” - very poor name for the variable, don’t use it for your code):

```
df$GradeClassF <- factor(df$GradeClass,
  levels = c(0,1,2,3,4),
  labels = c("A","B","C","D","F"))
```